MIL-P-55010A
16 February 1965

SUPERSEDING
MIL-P-55010(CE)
14 February 1958

MILITARY SPECIFICATION

PLASTIC SHEET, POLYETHYLENE TEREPHTHALATE

This specification has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a transparent, colorless, polyethylene terephthalate plastic in sheet form.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-268	- Paper, Kraft, Untreated, Wrapping.
UU-P-553	- Paper, Wrapping, Tissue.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-591	- Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).

FSC 9330

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STANDARDS

Federal

FED. TEST METHOD STD.- Plastics: Methods of Testing.
No. 406

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York. 10016.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Materials. The material shall consist of clear polyethylene terephthalate plastic sheet without the addition of dyes, pigment, fillers, or bonding agents.

3.2 Storage conditions. When tested as specified in 4.5.3, the material shall not be affected permanently by the temperatures and relative humidity conditions encountered in storage.

3.3 Dimensions.

3.3.1 Length and width. The material shall be furnished in the lengths and widths specified (see 6.2). A tolerance of plus or minus 1/8 inch of the length and width specified shall be allowed.

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3.3.2 Thickness. Unless otherwise specified (see 6.2), the thickness shall be 0.0075 inch, plus or minus 0.0008 inch, and the difference between the thickest and thinnest portions of any sheet shall not exceed 0.0006 inch.

3.4 Color. The material shall be free from opalescence and noticeable coloring when viewed at an angle of 90 degrees to the plane of the material. White incandescent, daylight-type fluorescent, or natural lighting incidence in the range of 10 to 90 degrees to the plane of the material shall be used for viewing purposes.

3.5 Transparency and optical properties. The material shall transmit not less than 90 percent of incident white light when tested as specified in 4.5.2. The material shall contain not more than eight defects per square yard, and the defective area shall be not more than 0.1 percent of the total area of the material. A defect is defined as any visible inclusion, discoloration, or spot larger than 1/16 inch in the greatest dimension.

3.6 Finish. The finished material shall have a smooth, polished surface on both sides.

3.7 Physical requirements. When tested as specified in 4.5.4, the material shall conform to the physical requirements shown in table I.

Table I. Physical Requirements.

	Maximum	Minimum
Specific gravity, 25° C. (77° F.)	1.45	1.35
Tensile strength, psi	---	16,000
Elongation at rupture, percent	---	60
Tear resistance, lb/in	---	600
Water absorption, 24-hour immersion, percent	0.7	---
Burning rate, inches per minute	12.0	---

3.7.1 Dimensional stability. The dimensional change of the material under changes in temperature and relative humidity shall be not more than 0.01 inch, and the differential dimensional change of the material shall be not more than 0.003 inch when tested as specified in 4.5.5 and 4.5.6.

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3.7.2 Flatness. The flatness factor of the finished material shall be not greater than 0.01 inch per length in inches when the material is tested as specified in 4.5.1. The material shall be free from noticeable ripple or waviness.

3.8 Workmanship. The plastic sheets shall be clean and free from dirt, grease, oil spots, or any harmful extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all material of the same type produced under like conditions in one unchanged process by one manufacturer.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.1.2 Sampling for tests. One-half of one percent of the lot, but not less than two sheets, shall be selected at random for testing.

4.4 Examination. Samples selected in accordance with 4.3.1.1 shall be examined for the following defects. Lot acceptance or rejection shall be based on an AQL of 1.5 percent defective for major defects and 6.5 percent defective for minor defects.

Major

101. Dimensions not as specified.
102. Scratches, cuts, holes, or tears.
103. Not smooth polished surface on both sides.
104. Noticeable opalescence or coloring.

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Minor

201. Greasy or oily film, dirt, foreign material, or imbedded particles.

202. Wrinkles, creases, or unevenness.

4.5 Tests. Samples selected in accordance with 4.3.1.2 shall be subjected to the following tests in serial order. Failure of a sample to pass any test shall be cause for rejection of the lot which the sample represents.

4.5.1 Flatness test. The sheets shall be placed on a plane surface and smoothed by hand, and the four corners shall be held down by weights not more than 4 pounds each and covering not more than 4 square inches of the area of each sheet. The maximum departure of the lower surface of the sheet from the plane surface shall be measured to the nearest 0.01 inch, and the flatness factor shall be determined by dividing the departure so measured by the length of the sheet in inches. If 5 percent of the sample fails to meet the requirements in 3.7.2, the sample shall have failed the test.

4.5.2 Light transmittance test. The percentage of white light transmitted shall be determined at not less than six points on each sample sheet by means of a photoelectric densitometer to determine compliance with the light transmittance specified in 3.5. If 3 percent of the points or one point, whichever is greater, examined fail to transmit the required percentage of light, the sample shall have failed the test.

4.5.3 Storage test. Not less than two sample sheets shall be subjected to a cycle of temperature and relative humidity extremes in the following order:

(a) A high-temperature series, consisting of three exposures of 4 hours each on 3 successive days at a temperature of 155° F. plus or minus 5° F. and a relative humidity of 15 percent plus or minus 5 percent.

(b) A low-temperature condition, consisting of 24 hours exposure at a temperature of minus 65° F. plus or minus 5° F.

(c) A high humidity condition, consisting of 24 hours exposure at a relative humidity of 98 percent plus or minus 2 percent and a temperature of 80° F. plus or minus 2° F.

(d) A normal temperature and humidity condition, consisting of 8 or more hours at a temperature of not less than 65° F. nor more than 85° F. and a relative humidity of not less

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than 20 percent nor more than 80 percent. Upon completion of the cycle, the sample shall be reexamined as specified in 4.4 and retested as specified in 4.5.1 and 4.5.2. Noncompliance with 3.2 shall constitute failure of this test.

4.5.4 Physical tests. The physical characteristics of the material shall be determined in accordance with FED. TEST METHOD STD. No. 406 for compliance with 3.7:

Characteristic	Method
Specific gravity	5011
Tensile strength, method A	1013
Elongation at rupture, method A	1013
Tear resistance	1121
Water absorption, procedure A	7031
Burning rate	2022

4.5.5 Thermal dimensional stability test.

4.5.5.1 Specimens. Two 3- by 12-inch specimens shall be cut from each sheet of the sample. One specimen shall be cut from the lengthwise dimension of the sheet and the other from the crosswise dimension of the sheet. The specimens shall be index marked at a distance of 10 inches, plus or minus 0.1 inch, between the marked points.

4.5.5.2 Procedure. The marked specimens shall be conditioned for not less than 24 hours at each of the following temperatures at a relative humidity of 50 percent plus or minus 2 percent: 60° F. plus or minus 1° F.; 110° F. plus or minus 1° F.; and 60° F. plus or minus 1° F. in the order given. The distance between the index marks shall be measured to within plus or minus 0.0005 inch at the end of each of the three 24-hour conditioning periods without removing the specimens from the conditioning temperatures. The difference between the average distance between the index marks measured at 60° F. and the distance between these marks measured at 110° F. shall be considered the thermal dimension change for determining compliance with the requirements of 3.7.1. The smaller difference so found for each pair of specimens from each test sheet shall be subtracted from the larger difference found for the pair

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to determine compliance with the differential dimension change requirement of 3.7.1. Noncompliance with 3.7.1 shall constitute failure of this test.

4.5.6 Hygroscopic dimensional stability test.

4.5.6.1 Specimens. Two 3- by 12-inch specimens shall be cut from each sheet of the sample. One specimen shall be cut from the lengthwise dimension of the sheet and the other from the crosswise dimension of the sheet. The specimens shall be index marked at a distance of 10 inches, plus or minus 0.1 inch, between the marked points.

4.5.6.2 Procedure. The marked specimens shall be conditioned for not less than 24 hours at each of the following relative humidities at a temperature of 70° F. plus or minus 1° F.: 30 percent plus or minus 2.0 percent; 80 percent plus or minus 2.0 percent; and 30 percent plus or minus 2.0 percent, in the order given. The distance between the index marks shall be measured to within plus or minus 0.0005 inch without removing the specimens from the conditioning relative humidities, at the end of each of the three 24-hour conditioning periods. The difference between the average distance between the index marks measured at 30 percent relative humidity and the distance between these marks measured at 80 percent relative humidity shall be considered the hygroscopic dimension change for determining compliance with 3.7.1. The smaller difference so found for each pair of specimens from each test sheet shall be subtracted from the larger difference found for the pair to determine compliance with the differential dimension change requirement of 3.7.1. Noncompliance with 3.7.1 shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined and tested for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The plastic sheets in quantities of 100 shall be interleaved with sheets of tissue paper conforming to UU-P-553 and wrapped with kraft wrapping paper conforming to UU-P-268, grade B, secured with tape conforming to PPP-T-76. The sheets shall be packaged in a box conforming to PPP-B-636, type I, class 2, compliance symbol as applicable, style FTC, in quantities not to exceed 500 sheets. The boxes shall be close fitting and shall be closed and sealed in accordance with the appendix to the box specification.

5.1.2 Level C. The sheets shall be packaged in accordance with the supplier's standard practice.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packaged sheets shall be packed in boxes conforming to PPP-B-621, class 2, style 2, 2-1/2, or 3, or PPP-B-601, overseas type, style A or B, plywood type III, class 1. The boxes shall be close fitting and shall be closed and strapped in accordance with the appendix to the applicable box specification, except that straps shall be flat steel, type I, class B. The gross weight of each box shall not exceed 200 pounds.

5.2.2 Level B. The packaged sheets shall be packed in boxes conforming to PPP-B-621, class 1, style as applicable; PPP-B-601, domestic type, style optional, plywood type IV; PPP-B-585, class 1 or 2, style optional; or PPP-B-591, domestic type, style optional. The boxes shall be close fitting and the gross weight of each box shall not exceed 200 pounds.

5.2.3 Level C. The packaged sheets shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The plastic material is intended for use in Military cartographic operations. The plastic sheet may be used for glazing which is exposed only intermittently to sunlight.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Length and width required (see 3.3.1).

(c) Thickness required if other than as specified in 3.3.2.

(d) Level of packaging and packing required (see 5.1 and 5.2).

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6.3 This material is available in 0.003-inch and 0.005-inch thicknesses in addition to that specified in 3.3.2. The thickness and square yards per pound are as follows:

Thickness in.	Weight sq yd/lb
0.003	5.13
0.005	3.09
0.0075	2.04

Custodians:

Army - MO(ERDL)

Navy - YD

Air Force - 69

Review activities:

Army - MO MD MR MU

Navy - YD

Air Force - 69

User activities:

Army - EL GL MI

Navy - CG MC

Air Force - None

Preparing activity:

Army - MO(ERDL)

Project No. 9330-0209

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U. S. ARMY MOBILITY COMMAND
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WARREN, MICHIGAN

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4861, WHICH IS OBSOLETE

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		$

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SUBMITTED BY (Printed or typed name and activity)	DATE

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